

# KEVIN GREIF (he/him)

ARTIFICIAL INTELLIGENCE — SCIENTIFIC RESEARCH — STATISTICAL METHODS

*Physics Ph.D. candidate with 5+ years of experience developing and deploying ML solutions for complex scientific problems. Published researcher specializing in statistical ML, generative models including diffusion, and large-scale HPC deployments. Seeking applied AI research roles in natural sciences or related fields.*

## EXPERIENCE

### Ph.D. candidate in Physics - AI methods development 2020 - Present

*University of California, Irvine*

- Developed novel **latent diffusion models** for inverse problems in particle physics, achieving state-of-the-art performance on high-dimensional scientific data
- Expertise in designing artificial intelligence systems for handling complex variable-dimensional data structures
- Published research in peer-reviewed conference proceedings (NeurIPS 2023, SciPost Physics 2024/2025)
- Presented research at international physics and machine learning conferences

### Researcher at CERN - AI methods deployment 2020 - Present

*ATLAS Collaboration*

- Trained thousands of large attention-based neural networks models on HPC resources
- Experience in developing and optimizing complicated neural network training workflows
- Benchmarked, optimized hyper-parameters, and quantified uncertainties for many neural network based classifiers. Improved performance by a factor of 5x over traditional methods
- Created and released open-source datasets and evaluation tools, facilitating reproducibility and community adoption

### Undergraduate Research Assistant - Neural Network Architecture Design 2018 - 2020

*University of Notre Dame*

- Designed physics-inspired neural network architectures incorporating domain knowledge, achieving improved performance over generic architectures

### Teaching, Leadership, and Service

- **ATLAS Early Career Scientist Board (2024-Present)**: Advocate for 1000+ early career scientists within international scientific collaboration, organize career development workshops and networking events
- **Teaching Assistant (2019-2021)**: Taught and mentored 100+ students in physics and computational methods courses

## SKILLS

**Artificial Intelligence:** Attention-based neural networks, diffusion models, hyperparameter optimization

**Tools and infrastructure:** PyTorch, PyTorch Lightning, TensorFlow, JAX, Git, Weights and Biases

**Programming:** Python, C++, Git, HPC resources and distributed computing, CI/CD pipelines, testing frameworks

**Data Science:** Large-scale data processing, statistical methods, data visualization

**Research:** Published author (NeurIPS, SciPost Physics), peer review, technical writing, scientific communication

**International Collaboration:** Active member of large (1000+ researchers) scientific collaboration, experience working in teams to produce high-quality research results

## EDUCATION

### University of California, Irvine 2020 - Present (Expected 2026)

Ph.D. in Physics, Department of Physics and Astronomy — Advisor: Daniel Whiteson

*Focus: Artificial Intelligence applications in particle physics research*

### University of Notre Dame 2016 - 2020

B.S. in Physics and Program of Liberal Studies, Magna Cum Laude

*Honors: Outstanding Physics Major Award, Departmental award for outstanding writing, Phi Beta Kappa*

## SELECTED PUBLICATIONS

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### **End-to-end latent variational diffusion models for inverse problems in high energy physics** — [NeurIPS](#)

Proposes a novel latent variational diffusion model. Applied to solving an important inverse problem in particle physics, demonstrating then state-of-the-art performance for generative model based approaches to this problem.

### **Full event particle-level unfolding with variable-length latent variational diffusion** — [SciPost Physics](#)

Proposes a latent variational diffusion model for solving variable dimensional inverse problems in particle physics. Main development was the ability to process variable-length feature spaces.

### **Generative unfolding of jets and their substructure** — [arXiv:2510.19906](#)

Proposes a flow-matching model based method for solving a very high (100+) and variable-dimensional inverse problem in particle physics.

### **Accuracy versus precision in boosted top tagging with the ATLAS detector** — [JINST](#)

Benchmarking of various methods for solving an important classification problem in particle physics. Also includes rigorous uncertainty quantification for the performance of the methods, and documents the release of a high-quality public dataset for the community.

### **The landscape of unfolding with machine learning** — [SciPost Physics](#)

A community comparison study of unfolding methods in particle physics. I curated one of the comparison datasets used in this study.